

Exercise 4

The maximum number of elements in a row of A is equal to the max number of triangles sharing node that represents that row. This makes the matrix A sparse.

Exercise 5

Property (iii) states that: if Σ_K is a set of n linear functionals $\{\sigma_i\}_{i=1,\dots,n}$ such that for any real scalars α_i , $i = 1, \dots, n$ there exist a unique function $p \in P_K$ that satisfies (unisolvence):

$$\sigma_i(p) = \alpha_i$$

Let (P) be the sentence: $\sigma_i(p) = \alpha_i \iff p = 0, i = 1, \dots, n$.

- (iii) $\implies$ (P):

Notice that

$$\sigma_i(0) = \sigma_i(0 + 0) = 2\sigma_i(0) \implies \sigma_i(0) = 0, \forall i$$

Using (iii), $p \equiv 0$ is the only function that satisfies $\sigma_i(p) = 0, \forall i$. So $\sigma_i(p) = 0 \implies p = 0$.

If $p = 0$, we already shown that $\sigma_i(p) = 0$. Therefore, if (iii) is valid, (P) holds.

- (P) $\implies$ (iii)

Take $\{\alpha_i\}_{i=1}^n$ scalars and $v, w \in P_K$ such that $\sigma_i(v) = \alpha_i$ and $\sigma_i(w) = \alpha_i$, $i = 1, \dots, n$. For each i , subtracting both equations:

$$\sigma_i(v) - \sigma_i(w) = \alpha_i - \alpha_i = 0 \implies \sigma_i(v - w) = 0, i = 1, \dots, n$$

As we're supposing (P), we have that $v - w = 0$. So $v = w$ and (iii) is true.

$\therefore$ (P) is equivalent to (iii).

Exercise 6

For each element of the triangulation, as we're using second degree polynomials in 2 dimensions, we need 6 degrees of freedom (DoF), located at the element's vertices and the midpoints of the edges. Notice that V_h is a space of continuous functions. So, the DoFs between elements are shared. Therefore, for each node and edge there is a degree of freedom (DoF). We can write:

$$\dim(V_h) = N_v + N_{edges}$$

Where N_v and N_{edges} are the numbers of vertices and edges of the mesh respectively.

Exercise 7

- Hermite

Uses $P_3(K)$. Output function is C^1 in the whole domain. Uses 10 DoFs: 3 at each node (2 associated with the derivatives) and one at the baricenter.

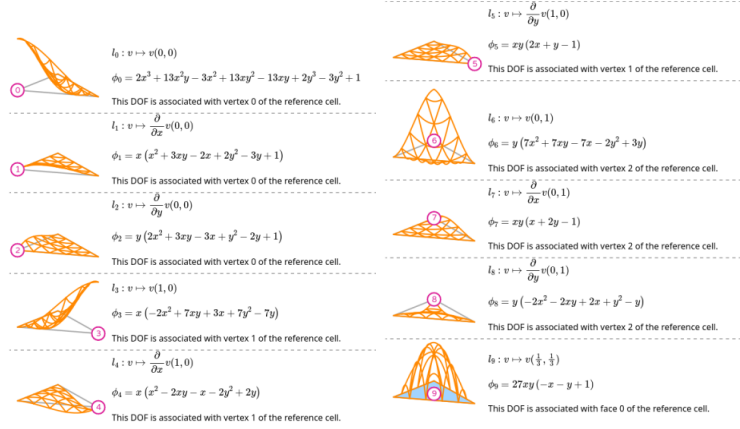


Figure 1: Basis functions of Hermite finite element

- Degree 1 Taylor

Continuous across elements. 3 DoFs at the baricenter, two of them regarding the derivative.

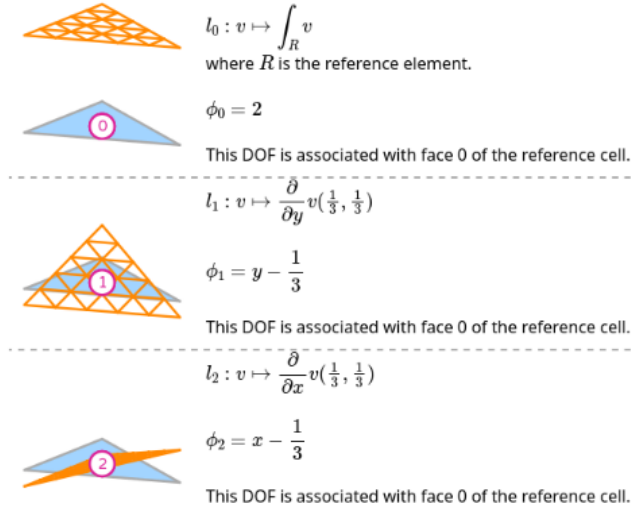


Figure 2: Basis functions of Taylor finite element

- Vector lagrange (Degree 1)

Describes vector functions ($f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$). It's continuous across elements and uses 6 DoFs, 2 at each node.

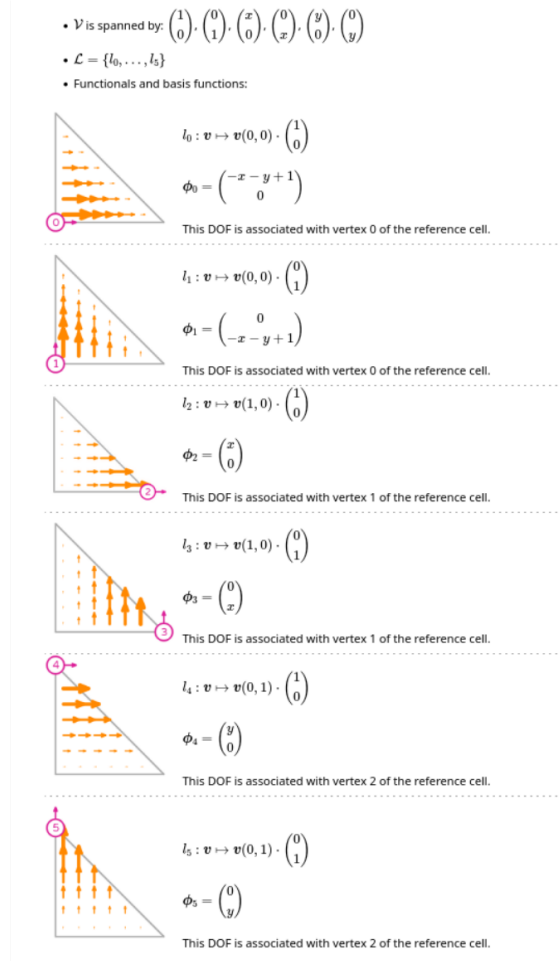
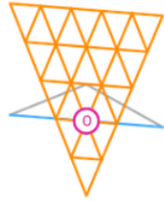


Figure 3: Vector Lagrange basis functions

- Crouzeix-Raviart

Discontinuous across elements. Uses 3 DoFs per element, 1 at the midpoint of each edge. At those points, there's continuity.

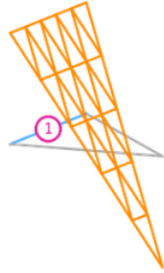
- $\mathcal{V}$ is spanned by: $1, x, y$
- $\mathcal{L} = \{l_0, \dots, l_2\}$
- Functionals and basis functions:



$$l_0 : v \mapsto v\left(\frac{1}{2}, 0\right)$$

$$\phi_0 = 1 - 2y$$

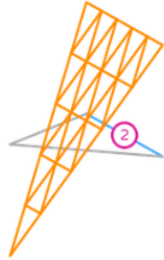
This DOF is associated with edge 0 of the reference cell.



$$l_1 : v \mapsto v\left(0, \frac{1}{2}\right)$$

$$\phi_1 = 1 - 2x$$

This DOF is associated with edge 1 of the reference cell.



$$l_2 : v \mapsto v\left(\frac{1}{2}, \frac{1}{2}\right)$$

$$\phi_2 = 2x + 2y - 1$$

This DOF is associated with edge 2 of the reference cell.

Figure 4: Basis functions of Crouzeix-Raviart

- Vector bubble enriched Lagrange (Degree 1)

Describes vector functions and is continuous across elements. Uses 8 DoFs: 2 for each node (1 for each direction) and 2 at the baricenter (1 for each direction).

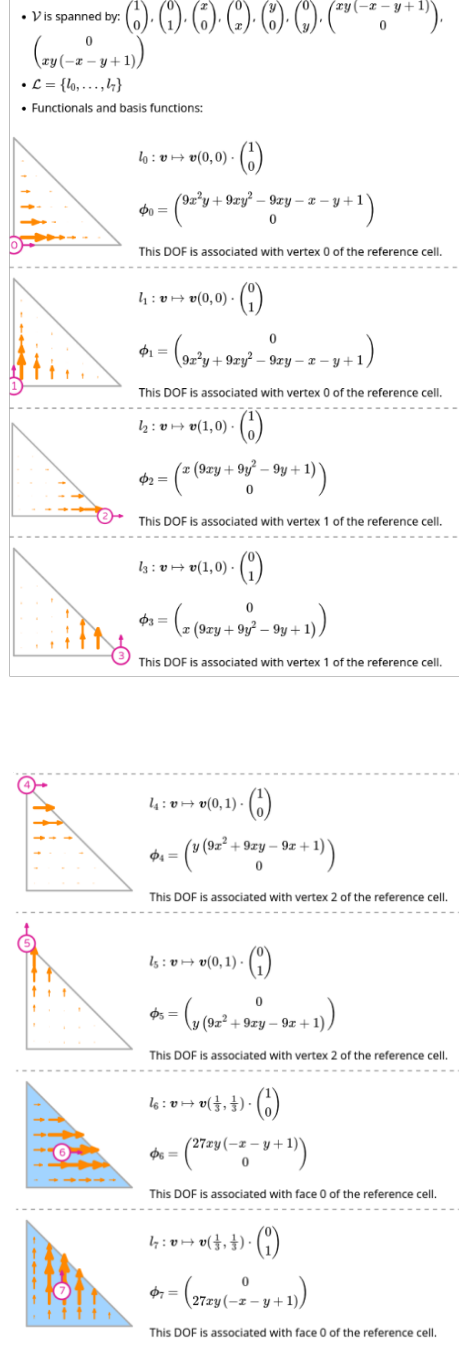


Figure 5: Basis functions of vector bubble enriched Lagrange